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Stents in Coronary Angioplasty: A Mechanical Review of Drug-Eluting Stent Design

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1 Introduction

Cardiovascular diseases are the leading cause of death globally, representing approximately 32% of all global deaths [1]. Coronary artery disease (CAD) is amongst the most common cardiovascular diseases and is caused by atherosclerosis, a condition characterised by the accumulation of lipids, cholesterol and fibrous elements within the tunica intima. The inflammatory response of the immune system leads to macrophages engulfing the lipids, forming a fatty streak of foam cells and collagen which develops into plaque buildup [2]. The rupture of this plaque triggers platelet aggregation causing vessel occlusion, restricting blood flow to cardiac tissue and inducing myocardial infarction risk. Given the mortality burden, mechanical and surgical treatment methods are crucial to ensuring longevity and improving patient well being. This essay reviews the engineering design and use of drug-eluting stents (DES) to restore lumen diameter, focusing on the required optimisation of mechanical integrity, material properties and drug release. Attention is given to the advantages and limitations of DES along with current innovations and improvements to stent design.

2 Stent Design

2.1 Categories of Stents

There are three main types of stents: Bare-metal stents (BMS), Drug-eluting stents (DES) and bioresorbable stents (BRS). First generation BMS (introduced in 1987 [3]) were made from 316L stainless steel or cobalt-chromium due to their corrosion resistance [4]. Whilst being an effective solution, due to their non-degradability, there is a significant increase in risk of postoperative complications [5] such as in-stent restenosis which requires repeat intervention in 12-20% of patients [6]. The development of DES addressed this issue by integrating a drug-eluting coating made from biocompatible polymers such as phosphorylcholine, polylactic acid, poly(vinylidencefluoride-co-hexafluoropropenen), and polyvinylpyrrolidone [7] and incorporating antibiotic drugs such as sirolimus or paclitaxel [8]. This incorporation of targeted drug release encourages re-endothelialisation (arterial

healing) and reduced the need for repeat procedures due to postoperative complications to about 5% [6]. BRS are a recent development in stent design, focusing on combating issues with permanent structures by utilising biodegradable polymers such as PLA and PLLA [7]. Recent BRS designs include the "Igaki-Tamai" stent, made from poly-L-lactic acid which takes between 18 and 24 months to biodegrade and resulted in a restenosis rate of 10% from 15 patients[8].

2.2 Performance Metrics

Coronary stents act as cylindrical load-bearing scaffolds and require a balance of radial strength, flexibility and fatigue resistance to survive more than 30 million cardiac cycles per year. There are many performance metrics [table 1] that assess the stents' ability to provide structural support with minimal vessel injury whilst minimising the surrounding blood flow environment [9].

2.3 Structural Configuration

The structure of DES is centred around struts, crowns and connectors which form ring or helical patterns that are then connected to one another[9]. These patterns are optimised to allow for uniform deformation during balloon expansion whilst minimising local stress concentrations at the crown and connector joints. It is also important to note that repeated patterns with geometric uniformity allow for uniform laminar flow and so irregular geometries aren't considered when designing stents. A recent study [10] utilised FEA through ANSYS software tested multiple stent structures derived from commercial stents and concluded that the structural pattern depicted in figure 1 provided the lowest residual stress of 120.46 MPa and a lower predicted percentage of thrombosis formation compared to the straight-link or hexagonal counterparts. The thickness of the struts also plays a vital role in determining the bending stiffness, radial force and fatigue life of the stent. The development of second-generation DES used cobalt, chromium and platinum alloys which allowed for the use of thinner struts of 80-90 μm compared to 132-140 μm used in first-generation DES [9]. These thinner strut designs are useful as they reduce arterial injury and improve deliverability, however, consideration of material

tensile strength and young's modulus must be taken into account as thinner struts increase peak stress and the risk of fatigue cracking.

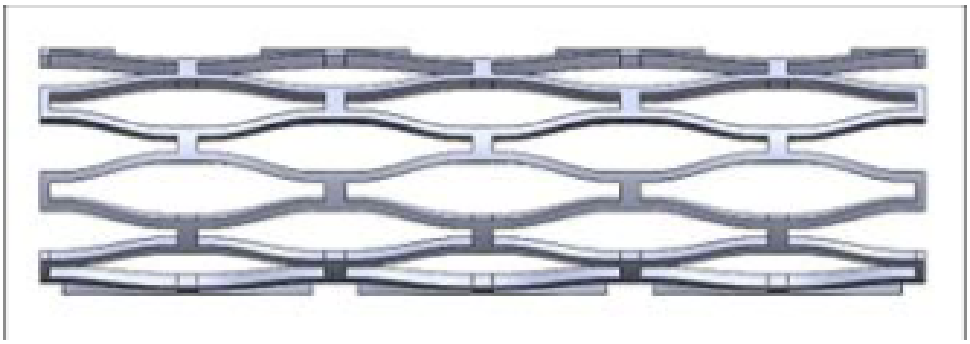


Figure 1: Typical geometry of an undeployed coronary artery stent.

2.4 Modelling and Load Analysis

The Bloch-Floquet technique can be used to analyse the behaviour of periodic systems under harmonic perturbations [11]. This is done by treating the stent as a periodic waveguide to identify resonant frequencies and obtaining deformation modes of a stented blood vessel as seen in figure 2.

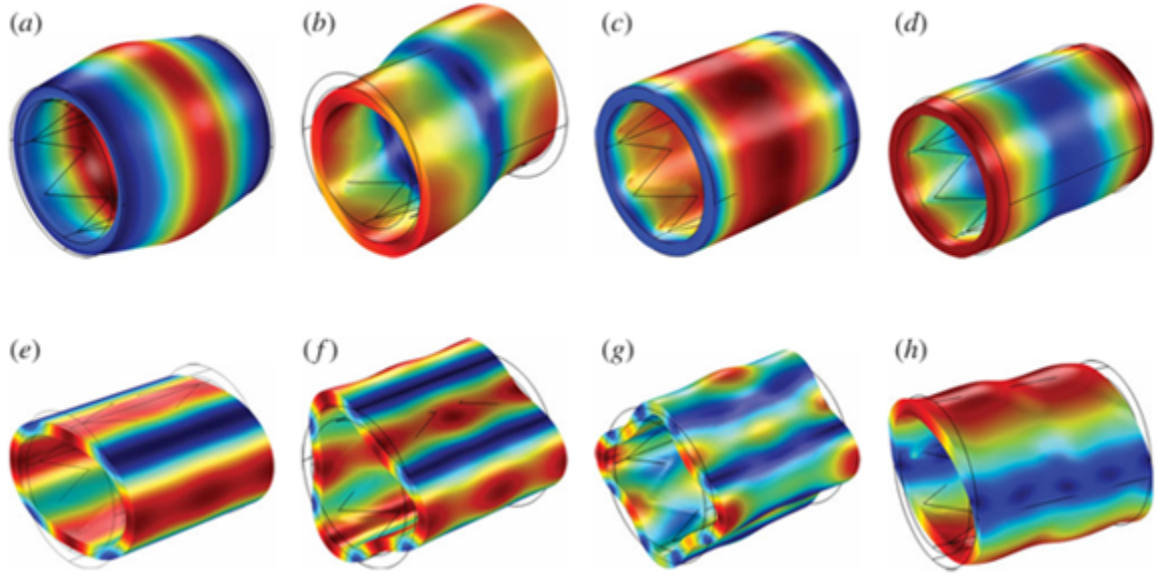


Figure 2: Wavenumber response and deformation modes of a stent: (a)-(d) Axisymmetric modes; (e)-(h) Flexural and torsional modes.

Once these frequencies are obtained, they can be linked to activities that produce these frequencies due to cyclic pumping which can have adverse effects on the stent. Mathematical models can also be employed to analyse the mechanical behaviour of stents. A

study in by J. Tambača, M. Kosor, S. Čanić, and D. Paniagua [12] presented a model to predict axial and radial displacement which showed that stents that are maximally expanded before implantation can withstand greater pressures and that stiffer stents deform less whilst nonuniform pressures cause higher localised deformation. The relationships obtained were:

$$\text{Longitudinal displacement} = \frac{k_L E_w t^4}{(RLu)}$$
$$\text{Radial displacement} = \frac{k_r E_w t^4}{(R^2 u)}$$

These FEA studies [12,10] also assess the Von Mises stress distribution during expansion, showing that optimised designs achieve uniform stress distribution between the crowns and connectors to minimise plastic deformation.

3 Drug Elution

The drug eluting layer is made up of a polymer containing low doses of sirolimus and then a thin layer of polymer that is drug-free to act as a barrier to diffusion and thus allows for gradual elution of the drug. The targeting occurs as sirolimus requires a binding protein (FKBP) found only on smooth muscle cells and lymphocytes[8].

The 2013 study by Sean McGinty et al [13] considered analytical solutions to model drug release concentrations by modelling the interface between the polymer and the tissue as a coupled system, satisfying a Volterra integral which can be solved to find the mass of drug required. The Glerkin Scheme [14] can also be used to solve a 2D model of drug release to optimise drug concentrations and targeted release.

4 Conclusion

DES are a milestone in biomedical engineering, showcasing how structural optimisation and material selection can solve complex biological challenges such as coronary artery disease. Thin cobalt-chromium struts used in repeated, flexible ring geometries

provide the best balance between radial strength and axial flexibility, reducing deployment defects such as radial recoil and dogboning [Table 1] whilst maintaining mechanical endurance under millions of cardiac cycles and biocompatible polymer coatings provide targeted pharmacological control to further reduce thrombosis. Whilst advanced manufacturing, computational optimisation and mathematical analysis provide advanced and effective designs, there are still limitations to DES such as fatigue-induced fracture and incomplete biodegradability. Future progress relies on incorporating biodegradable metallic alloys to enhance biocompatibility which is at the forefront of stent design with the creation of BRS which aim to eliminate in-stent restenosis and thrombosis entirely.

A Appendix: Stent Performance Metrics

Table 1: Summary of Structural Intra-deployment Stent Performance Metrics.

Metric type	Performance Measure	Definition	Numerical Formula
Structural Intra-deployment	Stent recoil (%)	Reduction of stent diameter after balloon removal during deployment	$\left(1 - \frac{Diameter_{final}}{Diameter_{inflated}}\right) \times 100$
	Arterial stresses (structural)	Volume average stress developed in the artery during stent deployment	$\frac{\int_V \sigma dV}{\int_V dV}$
	Bending flexibility (crimped)	Ability of crimped stent to undergo bending deformation in a tortuous vessel path	$\int_0^{\kappa_l} M d\kappa$
	Stent stresses (structural)	Structural stress imparted on the stent during deployment	$\frac{\int_V \sigma dV}{\int_V dV}$
	Longitudinal resistance	Ability of deployed stent to resist deformation under an axial compressive load	$\int_0^{\gamma_d} F(\gamma) d\gamma$
	Dogboning	Slower expansion of stent middle as compared to end locations during deployment	$\frac{r_{distal} - r_{central}}{r_{distal}}$
	Foreshortening (%)	Decrease in axial length of stent during expansion	$\left(\frac{L_{final} - L_{initial}}{L_{initial}}\right) \times 100$

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